```
import numpy as np
import pandas as pd
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix,classification_report

X,y=make_classification(n_samples=400,n_features=2,n_informative=2,n_r
edundant=0,n_clusters_per_class=1,random_state=42)
df=pd.DataFrame(X,columns=["Age_Feature","Salary_Feature"])
df["Purchased"]=y
print(df.head())
features=df[["Age_Feature","Salary_Feature"]].values
label=df["Purchased"].values
x_train,x_test,y_train,y_test=train_test_split(features,label,test_siz
e=0.2,random_state=42)
model=LogisticRegression()
model.fit(x_train,y_train)
print("Training Score:",model.score(x_train,y_train))
print("Testing Score:",model.score(x_test,y_test))
predictions=model.predict(x_test)
print("Confusion Matrix:\n",confusion_matrix(y_test,predictions))
print("Classification Report:\
n",classification_report(y_test,predictions))
print("Predicted Purchase:",model.predict(np.array([[0.5,-0.5]]))[0])
```

```
   Age_Feature  Salary_Feature  Purchased
0     1.431868       -0.953095          0
1     0.323473       -1.833522          0
2     0.977841        1.501977          1
3     1.986525        0.602109          0
4     1.356117        0.280041          1
Training Score: 0.88125
Testing Score: 0.9
Confusion Matrix:
 [[34  7]
 [ 1 38]]
Classification Report:
               precision    recall  f1-score   support

           0       0.97      0.83      0.89        41
           1       0.84      0.97      0.90        39

    accuracy                           0.90        80
   macro avg       0.91      0.90      0.90        80
weighted avg       0.91      0.90      0.90        80

Predicted Purchase: 0
```